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SUPERSEDING
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MILITARY SPECIFICATION

SPLICING, CONDUCTOR, SOLDERLESS CONNECTIONS, PROCEDURE FOR

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers methods for splicing conductors using solderless, preinsulated, crimp type conductor splice.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace, and Control Systems, Procedures for

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STANDARDS

Military

MIL-STD-202	Test Methods for Electronic and Electrical Parts
MS25037	Crimp Tool, Hand, for Copper Insulated Terminals

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the Government or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Parts and materials. Unless otherwise specified on the engineering drawings, parts and materials utilized in the procedures described in this specification shall conform to the requirements specified herein. Where alternate materials are provided in the technical data package, it shall be the responsibility of the manufacturer to assure compatibility with all other materials and components of the assembly.

3.2 Operator training. Operators shall be trained in the procedures and techniques for fabricating crimped terminations to meet specification requirements herein.

3.3. Crimping tools. Standard crimping tools shall conform to MS25037 and this specification. Crimping tools not conforming to above standards shall be approved by the procuring activity prior to production use. The manufacturer of nonstandard tools shall furnish crimping die dimensions (indentor and nest) and "closed position" die clearance (opening between crimping dies). The manufacturer shall be responsible for the use of suitable gages or test methods, approved by the procuring activity, to determine proper tool adjustment.

3.4 Crimp tool inspection. Crimping tools used shall be checked prior to the beginning of production and at periodic intervals as specified in 4.5. The agency performing

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the crimping operations shall be responsible for maintaining adequate tool usage and tool inspection records satisfactory to the procuring activity to determine the acceptability of the crimping tools. Inspection records shall be kept complete and available to the Government at all times.

3.5 Performance. Conductor splices shall conform to the performance requirements of this specification.

3.5.1 Corrosion resistance. The crimped termination shall show no degradation in electrical requirements when subjected to a 5-percent salt solution in accordance with Method 101, Condition A, of MIL-STD-202.

3.5.2 Voltage drop. The voltage drop across the crimped connection shall be not greater than 5 millivolts at 1 ampere before and after 5 cycles of temperature cycling between 257 degrees Fahrenheit (F) and minus 85 degrees F.

3.5.3 Tensile strength. The crimped connection shall not break or become distorted to the extent it is unfit for the purpose intended when subjected to the tensile load specified in Table I.

3.5.4 Dielectric strength. The insulation of the crimped connection shall have a minimum dielectric strength sufficient to withstand 1,500 volts direct current (vdc) for not less than 1 minute between the crimped connection and any other separate circuit in the wire bundle. Leakage shall be not greater than 7 microamperes.

Table I

Tensile strength requirements	
Wire size designation, specification	Tensile strength (pounds)
AWG-26	7
AWG-24	10

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Table I (Continued)

AWG-22	15
AWG-20	19
AWG-18	38
AWG-16	50
AWG-14	70
AWG-12	110
AWG-10	180
AWG-8	225
AWG-6	300
AWG-4	400
AWG-2	550
AWG-1	650
AWG-0	700
AWG-00	750
AWG-000	825
AWG-0000	875

3.6 Preproduction sample. The sample for preproduction approval, if required, shall consist of the number of samples specified in the contract or order (see 6.2). Each preproduction sample shall consist of a 12-inch minimim representati section of the cable or harness with spliced conductors. This sample shall be in addition to the preproduction sample of the finished assembly, but may be concurrent with shield

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terminating and dead ending samples. The preproduction sample shall be terminated in accordance with applicable engineering drawings of one of the applicable cable ends. When more than one preproduction sample is specified, conductors shall be spliced representing each of the cable ends using such connections. Duplicate samples shall not be furnished unless the number of preproduction samples exceeds the number of applicable cable ends shown for the finished assembly. Each of the samples furnished shall be identified as to which of the ends of the cable assembly the sample represents. Each preproduction sample shall meet all requirements of the engineering drawing(s) and this specification except that connectors shall not be required.

3.7 Procedures. The procedures contained in this specification apply to splicing conductors with one piece insulated conductor splices.

a. Remove the primary conductor insulation as necessary to meet the provisions of (d) below, using procedures specified in MIL-S-45743. Jumper wire routed to solder connections or crimp pins on the connector shall be treated as primary conductors when connected to the connector. The length of the jumper shall be the minimum necessary and shall be of the type and size specified in the engineering drawings.

b. Select a crimping tool to meet the requirements specified herein (see 3.3 and 3.4).

c. Place the conductor splice into the crimping position in the crimping tool and squeeze handles to firmly hold the conductor splice without indenting the surface.

d. Insert all wires to be terminated in a given conductor splice to a depth such that the end of the primary wire insulation is within 1/16 inch of the end of the conductor crimp barrel with the conductor bottomed.

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e. Close the tool until the ratchet releases. For tools not equipped with ratchet release, squeeze tool handles until the dies are completely bottomed (dies closed to minimum opening).

f. Remove the tool and inspect the assembly for proper crimping.

3.8 Workmanship. Terminations completed under this specification shall be in accordance with all the requirements herein and shall be free from any defect which may affect its serviceability. Workmanship during termination shall be of the highest quality. The termination shall not exhibit splits or punctured insulation. Primary insulation, not including its jacket, shall not be cut, nicked, melted, or degraded. Wire marking shall not be obliterated or damaged beyond acceptance limits of the applicable wire and/or marking specification. The termination shall be free of foreign materials and contaminants.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspections shall be classified as follows:

a. Preproduction inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Crimping tool inspection (see 4.5).

4.3 Preproduction inspection. Preproduction inspection, if required (see 6.2), shall consist of all examinations and

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tests specified herein. Production approval will not be granted until the preproduction sample has been accepted.

4.3.1 Preproduction sample. The sample for preproduction approval, if required, shall consist of the number of samples specified in the contract or order (see 6.2). Each preproduction sample shall consist of a 12-inch minimum representative section of cable or harness used in assembly of the finished assembly with wires terminated. This sample shall be in addition to the preproduction sample of the finished assembly but may be concurrent with a preproduction sample of shield terminations. The preproduction sample shall be terminated (wired) in accordance with the applicable engineering drawings of one of the applicable cable ends. When more than one preproduction sample is specified, wires shall be terminated representative of each of the cable ends. Duplicate samples shall not be furnished unless the number of preproduction samples exceeds the number of applicable cable ends shown for the finished assembly. Each of the samples furnished shall be identified as to which of the ends of the cable assembly the sample represents. Each preproduction sample shall meet all requirements of the applicable engineering drawing(s) and this specification, except that connectors shall not be required.

4.3.2 Rejection. If any termination fails any examination specified herein, the preproduction sample shall be rejected and production approval will not be granted. Before a new preproduction sample is submitted, the manufacturer shall submit a detailed report covering the previous rejection and the action taken to prevent recurrence of the failure.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination specified in 4.6.1 and the tensile strength test of 4.6.5.

4.4.1 Quality conformance sample for examination. Unless otherwise specified, the quality conformance sample for the examination of 4.6.1 shall be one hundred percent of production terminations.

4.4.2 Quality conformance sample for tests. Unless otherwise specified, the sample for the test of 4.6.5 shall

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consist of prepared samples equal to a minimum of four percent of production terminations but not less than two samples for any one lot. The samples shall be prepared during regular production crimping utilizing the same tools and procedures. Each termination sample shall be representative of each size of termination of the lot in production. Each sample shall be approximately 12 inches long. Upon satisfactory exhibition of high quality performance and written request by the manufacturer to the procuring activity, a reduced sampling plan may be approved.

4.5 Crimping tool inspection. Prior to initial production operation and after each 40 hours of production use or each 30 days whichever comes first, the crimping tool die clearance with the tool in the fully closed position, shall be gaged with "GO" and "NO GO" gages. Gages for acceptable tools shall be as recommended by the tool manufacturer and approved by the procuring activity.

4.6 Examination and tests.

4.6.1 Examination. An in process examination of each cable assembly shall be conducted to ensure compliance with the requirements of this specification, applicable standards, and drawings with respect to materials, workmanship, markings and dimensions, and to determine compliance with the assembly drawings (running lists, etc.).

4.6.2 Voltage drop. Millivolt drop across the crimped connection shall be measured between jumper wires. Test current shall be one ampere. The voltage drop shall not exceed 5 millivolts.

4.6.3 Temperature cycling. A minimum of one sample crimped to a conductor in accordance with the procedure of 3.7, as applicable, shall be subjected to a temperature cycling test by alternately placing the test termination in hot and cold temperature chambers for 30 minute periods. The temperature of the hot chamber shall be plus 257 degrees F (minimum) and the temperature of the cold chamber shall be minus 85 degrees F (minimum). During temperature cycling, the test sample shall be placed in room ambient temperature of 70 (plus or minus 5) degrees F for a minimum of 15 minutes between each hot and cold cycle of the test. After five cycles from hot to cold temperature the voltage drop across

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the connection shall be measured to determine that the crimped termination meets the voltage drop requirements of 3.5.2.

4.6.4 Corrosion resistance. A minimum of one crimped termination shall be tested for voltage drop prior to subjection to corrosion tests. The test sample shall be subjected to a 5-percent salt solution fog in accordance with 3.5.1. The test specimen shall be placed in a position within the test chamber which will permit the most uniform coverage of the test specimen by the fog. Following the salt spray (fog) test, the samples shall be washed thoroughly with distilled water and air dried for a minimum of 1 hour. The test specimen shall be subjected to the voltage drop test and the results compared to the original voltage drop tests to determine damage caused from salt spray exposure.

4.6.5 Tensile strength. The test sample shall be placed in a standard tensile testing machine and sufficient force applied to pull the conductor out of the conductor splice or break the wire. The speed of the holding head shall be not greater than 1-inch per minute. The conductor separation strength shall be greater than that specified in 3.5.3.

4.6.6 Dielectric strength. The dielectric strength shall be measured between each circuit terminated in accordanc with this specification and all other unconnected circuits in the wire bundle. A voltage of 1,500 vdc shall be applied to the test points. Leakage current shall not exceed 7 microamperes (see 3.5.4).

5. PREPARATION FOR DELIVERY

This section is not applicable to this specification.

6. NOTES

6.1 Intended use. The materials, tools, and procedures covered by this specification are intended for use in the fabrication of crimp type (solderless) terminations in single or multiconductor electrical circuits used in electrical systems of missile weapon systems.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction approval is required.

c. Number of units required for the preproduction sample (see 3.6 and 4.3.1).

d. Where the preproduction approval tests are to be conducted.

6.3 Definitions.

6.3.1 Conductor splice. For the purpose of this specification, the term conductor splice is defined as an insulated device used to electrically join and strengthen electrical conductors at a common termination point by crimping.

6.3.2 Crimp tool. For the purpose of this specification, crimp tools are defined as tools utilized to form solderless terminations or connections in electrical circuits.

6.3.3 Crimp connections. For the purpose of this specification, crimp connections are defined as those connections made by crimping conductor splices to conductors.

6.3.4 Nest. The nest of the crimp tool is the die that holds or anchors the conductor splice during crimping operations.

6.3.5 Indentor. The indentor of a crimp tool is the die making the indentation or crimp forming the connection of a conductor(s) to the conductor splice.

6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-16018, dated 25 October 1968.

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